

CONCEPT CHECK 9.3

- 1. VISUAL SKILLS** In the citric acid cycle shown in Figure 9.11, what molecules capture energy from the redox reactions? How is ATP produced?
- 2.** What processes in your cells produce the CO_2 that you exhale?
- 3. VISUAL SKILLS** The conversions shown in Figure 9.9 and step 4 of Figure 9.11 are each catalyzed by a large multi-enzyme complex. What similarities are there in the reactions that occur in these two cases?

For suggested answers, see Appendix A.

CONCEPT 9.4

During oxidative phosphorylation, chemiosmosis couples electron transport to ATP synthesis

Our main objective in this chapter is to learn how cells harvest the energy of glucose and other nutrients in food to make ATP. But the metabolic components of respiration we have examined so far, glycolysis and the citric acid cycle, produce only 4 ATP molecules per glucose molecule, all by substrate-level phosphorylation: 2 net ATP from glycolysis and 2 ATP from the citric acid cycle. At this point, molecules of NADH (and FADH_2) account for most of the energy extracted from each glucose molecule. These electron escorts link glycolysis and the citric acid cycle to the machinery of oxidative phosphorylation, which uses energy released by the electron transport chain to power ATP synthesis. In this section, you will learn first how the electron transport chain works and then how electron flow down the chain is coupled to ATP synthesis.

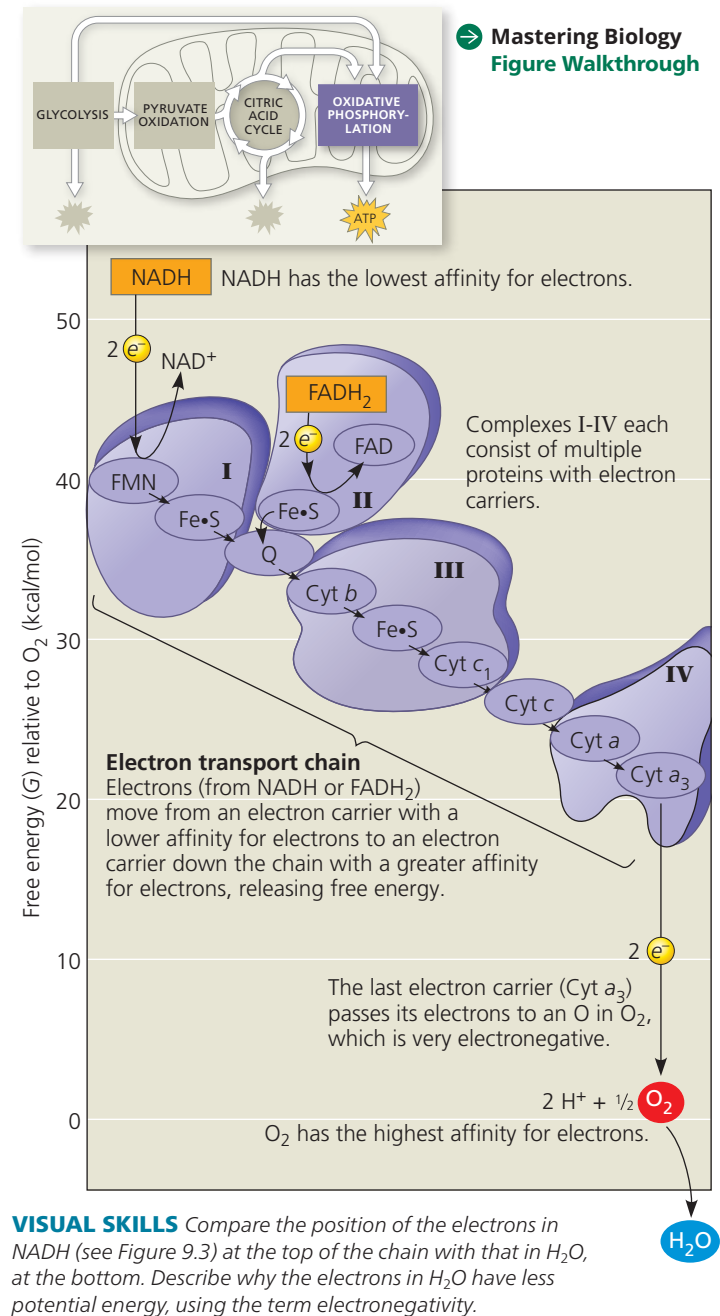
The Pathway of Electron Transport

The electron transport chain is a collection of molecules embedded in the inner membrane of the mitochondrion in eukaryotic cells. (In prokaryotes, these molecules reside in the plasma membrane.) The folding of the inner membrane to form cristae increases its surface area, providing space for thousands of copies of each component of the electron transport chain in a mitochondrion. Structure fits function: The infolded membrane with its concentration of electron carrier molecules is well-suited for the series of sequential redox reactions that take place along the electron transport chain. Most components of the chain are proteins, which exist in multiprotein complexes numbered I through IV. Tightly bound to these proteins are *prosthetic groups*, nonprotein components such as cofactors and coenzymes essential for the catalytic functions of certain enzymes.

Figure 9.12 shows the sequence of electron carriers in the electron transport chain and the drop in free energy as electrons travel down the chain. During this electron transport, electron carriers alternate between reduced and oxidized states as they accept and then donate electrons. Each component

▼ Figure 9.12 Free-energy change during electron transport.

The overall energy drop (ΔG) for electrons traveling from NADH to oxygen is 53 kcal/mol, but this “fall” is broken up into a series of smaller steps by the electron transport chain. (An oxygen atom is represented here as $\frac{1}{2} \text{O}_2$ to show that O_2 is reduced, not individual oxygen atoms.) To view these proteins in their cellular context, see Figure 6.32b.



VISUAL SKILLS Compare the position of the electrons in NADH (see Figure 9.3) at the top of the chain with that in H_2O , at the bottom. Describe why the electrons in H_2O have less potential energy, using the term electronegativity.

of the chain becomes reduced when it accepts electrons from its “uphill” neighbor, which has a lower affinity for electrons. It then returns to its oxidized form as it passes electrons to its “downhill” neighbor, which has a higher affinity for electrons.

Now let’s take a closer look at the electrons as they drop in energy level, passing through the components of the electron transport chain in Figure 9.12. We’ll first look at the passage of